

Model and Exact Answers for Final Project Quiz 3

Comparing the Information Gain of Alternative Data and Models

Note that sections 3.1 and 3.3 are model answers based on Professor Egger's model – there is no exact answer for these questions. Your answers will depend on your model. Section 3.2 does have the exact answers for the questions on the quiz. These are based on the Eggertopia data.

Requires Completion of Module 4 on Information Measures

3.0 Introduction

You realize that one way to characterize your binary classification model is as a tool to use current applicant data to reduce uncertainty about future default. Your model identifies dependencies between distributions in the six fields of data collected by the bank, and the probability distribution of eventual defaults. If we assume that the dependencies identified on the test set are stable and representative of all future data (a big assumption) we can draw some further conclusions about how much information gain, or reduction in uncertainty, is provided by your model at the chosen threshold.

3.1 Information Gain of Your Model over the Base Rate

The population of potential credit card customers consists of 25% future defaulters.

The base rate incidence of default (.25, .75) has an uncertainty, or entropy, of $H(.25, .75) = .25 \cdot \log_2 4 + .75 \cdot \log_2 1.333 = .81128$ bits.

Q: On your test set results, what is the conditional entropy of default, conditional on your test classifications? Hint: Refer back to the Test Incidence, Positive Predictive Value and Negative Predictive Value of your Test from Part 1 *[no exact answer]*

.7199 bits

Recall that conditional entropy is written $H(X|Y)$ and is determined as (the test incidence, or probability of a positive classification)* $H(+, -| Y = \text{Test is Positive})$ +(the probability of a negative classification)* $H(+, -| Y = \text{Test is Negative})$

You have also learned that the entropy of the original base rate, minus the conditional entropy of default given the classification test results, equals the Mutual Information between default and the test.

$$H(X) - H(X|Y) = I(X;Y).$$

Q: On your test set results, what is the Mutual Information, or information Gain, in average bits per event (information gain is calculated in units of bits per event)? *[no exact answer]*

.81128 -.7199 =

.0914 bits

Q: What is the Percentage Information Gain (PIG) of your test? *[no exact answer]*

.0914/.8113 =

11.3%

Q: How many dollars does the bank save, for every bit of information gain achieved by your model? *[no exact answer]*

Dollars saved are \$338/test

Bits extracted are .0914

Value would be =\$3,698/bit

3.2 Information Gain of Eggertopia Scores over the Base Rate

[For questions in this section, assume your model and the data it uses are not available – the bank's choice is between Eggertopia and the base rate].

Q: On the test set, What is the information gain, in average bits per event, over the base rate of (.25, .75) offered by the Eggertopia Scores?*

.1205 bits per event

Q: On the test set, what is the Eggertopia scores' Percentage Information Gain (PIG)?*

.1205/.8113=

14.9%

Q: If Eggertopia data were free, and your model were unavailable, what would the dollar savings per bit of information extracted be?*

\$413 saved

.1205 bits extracted

Value would be \$3,427 per bit.

3.3 Incremental Information Gain of Eggertopia Scores Compared to Your Model and Available Data

(For this section, assume your Model and the Data it uses *are* available).

Q: What is the *incremental* information gain of the Eggertopia scores, over your model, in average bits per event, if any? *[no exact answer]*

.1205 - .0914 =

.0291 bits/event

Q: What is the maximum (break-even) price the bank should pay for Eggertopia scores, per score, if your model and data are already available? *[no exact answer]*

\$75

Q: At the above maximum (break-even) price per score, what would be the value per bit of incremental information gained from the Eggertopia scores? Give your answer in \$/bit. *[no exact answer]*

\$75/.0291 =

\$2,576 per bit